

RAG System Architecture

Recursive splitting uses separators such as paragraphs, lines, and spaces to break texts into chunks. Semantic chunking uses word co-occurrence to break texts into chunks.

A cross-encoder reranker is powerful but expensive. It is usually applied only after a fast first-stage retrieval using a bi-encoder.

HyDE (Hypothetical Document Embeddings) creates a synthetic answer and embeds that instead of the real question.

ColBERT stores per-token vectors and uses late interaction scoring, which improves fidelity but increases memory usage.

Hybrid retrieval combines dense and sparse signals. Sparse retrieval is especially useful for exact matches.